

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 0.3698185962889105, median 0.3333620403244244, std: 0.2305819197970349
Gyroscope error (imu0): mean 2.586936040784385, median 1.8697767865058663, std: 3.9248990987173413
Accelerometer error (imu0): mean 4.309956761736137, median 2.6946151371477756, std: 8.411567335159468

Residuals

Reprojection error (cam0) [px]: mean 0.3698185962889105, median 0.3333620403244244, std: 0.2305819197970349
Gyroscope error (imu0) [rad/s]: mean 0.001483976417527654, median 0.0010725834011628777, std: 0.0022514888701716967
Accelerometer error (imu0) [m/s²]: mean 0.030536646815693125, median 0.019091725345791236, std: 0.05959713173009116

Transformation (cam0):

T_ci: (imu0 to cam0):

```
[ [ 0.01901582 -0.99900039 -0.04045509 0.06541471]
  [ 0.07258558 0.04173503 -0.9964886 -0.00460138]
  [ 0.99718089 0.01601259 0.07330665 -0.08450638]
  [ 0. 0. 0. 1. ] ]
```

T_ic: (cam0 to imu0):

```
[ [ 0.01901582 0.07258558 0.99718089 0.08335823]
  [-0.99900039 0.04173503 0.01601259 0.06689453]
  [-0.04045509 -0.9964886 0.07330665 0.00425601]
  [ 0. 0. 0. 1. ] ]
```

timeshift cam0 to imu0: [s] ($t_{imu} = t_{cam} + \text{shift}$)

0.0007692853107837673

Gravity vector in target coords: [m/s²]

[0.04168554 1.04657334 -9.75045483]

Calibration configuration

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cam0

Camera model: pinhole

Focal length: [411.75046101221477, 406.6683176274347]

Principal point: [289.2686675707598, 209.61441876821763]

Distortion model: radtan

Distortion coefficients: [0.011271099689313236, 0.001388004643421515, -0.015119164107067987, -0.017282641639372237]

Type: aprilgrid

Tags:

Rows: 6

Cols: 6

Size: 0.021 [m]

Spacing 0.0059997 [m]

IMU configuration

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IMU0:

Model: calibrated

Update rate: 50.0

Accelerometer:

Noise density: 0.0010019901002151698

Noise density (discrete): 0.007085139945439349

Random walk: 1.9893785362909353e-05

Gyroscope:

Noise density: 8.112529814510266e-05

Noise density (discrete): 0.0005736424844418254

Random walk: 6.28721982800146e-07

T_ib (imu0 to imu0)

[[1. 0. 0. 0.]

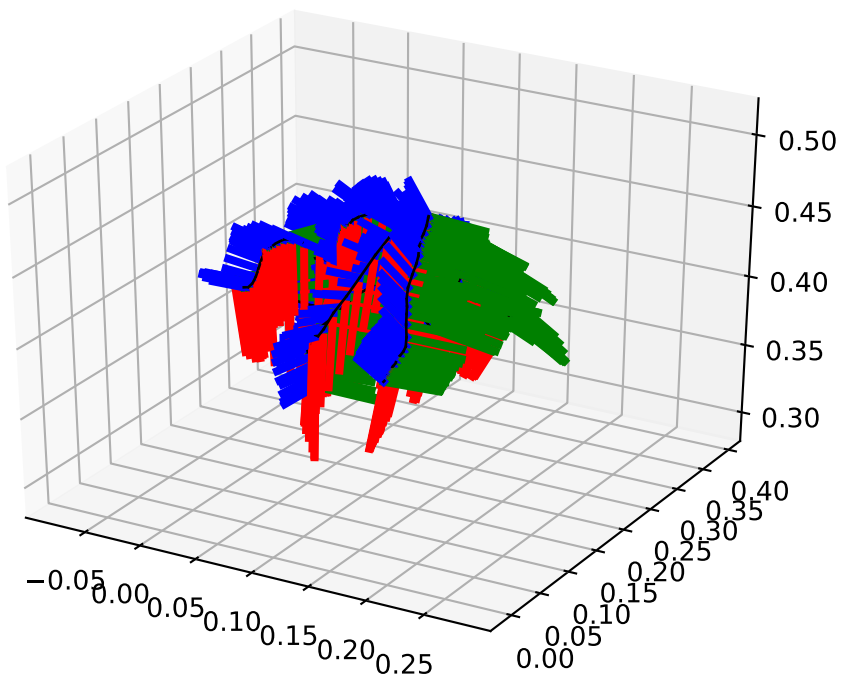
[0. 1. 0. 0.]

[0. 0. 1. 0.]

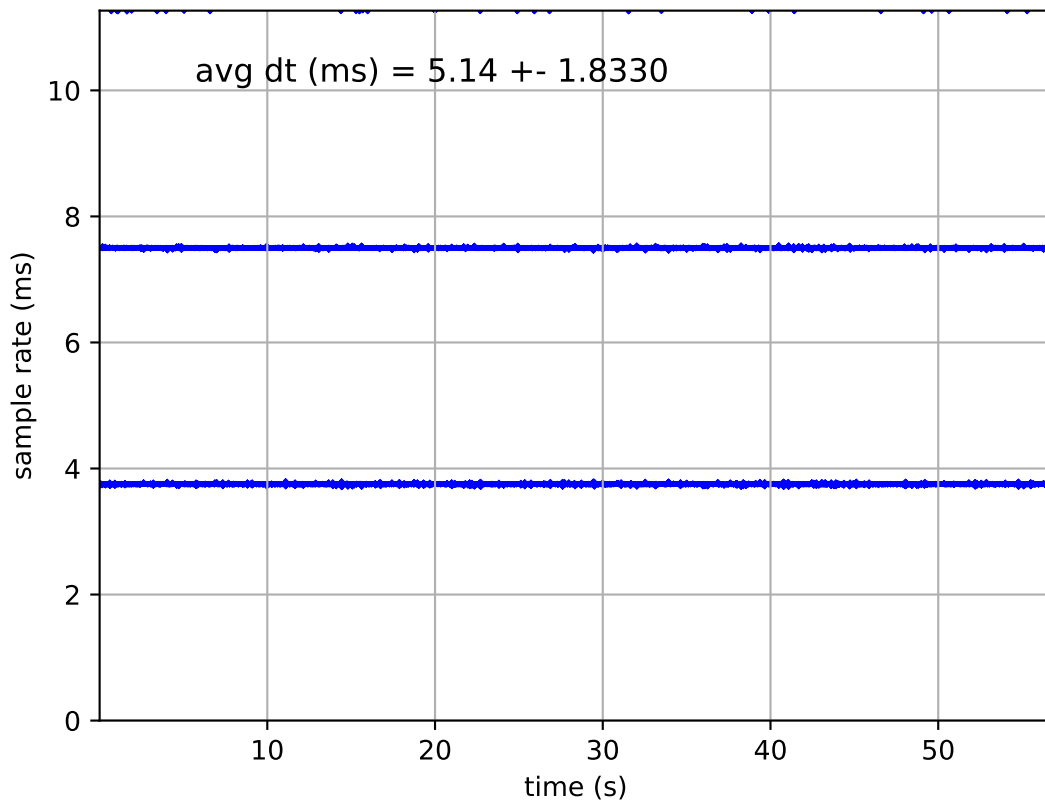
[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

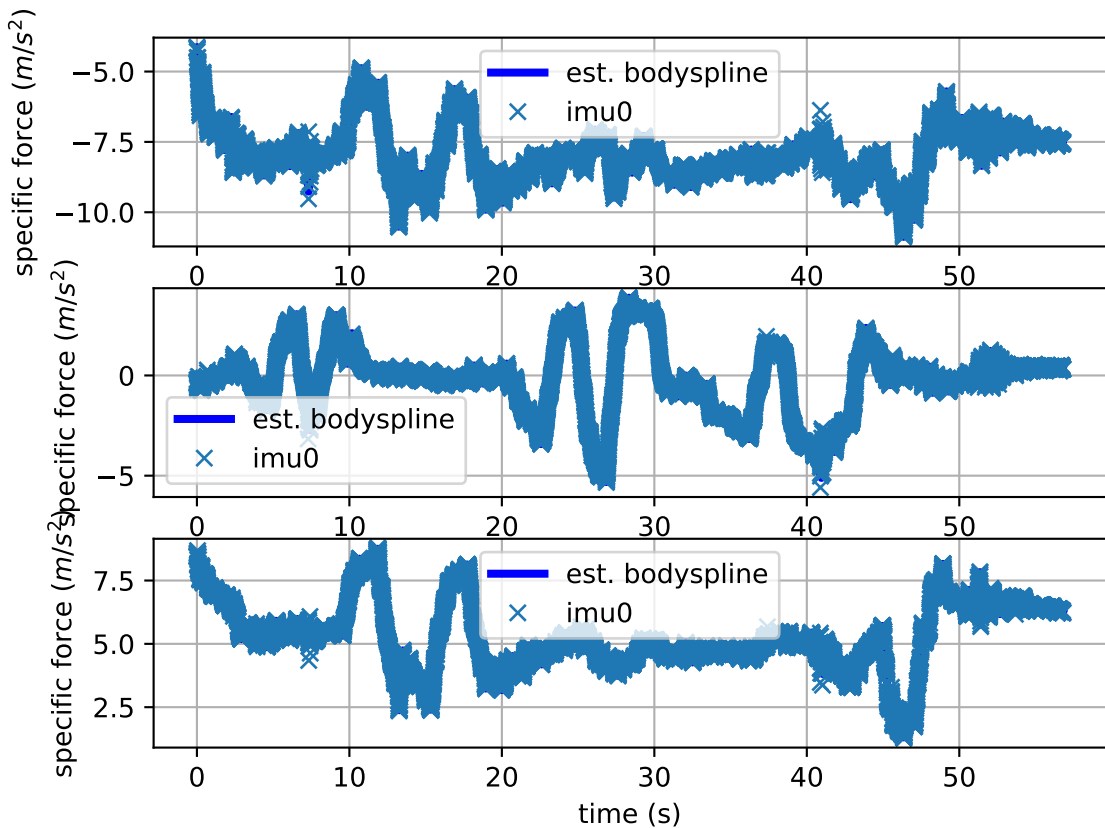
imu0: estimated poses



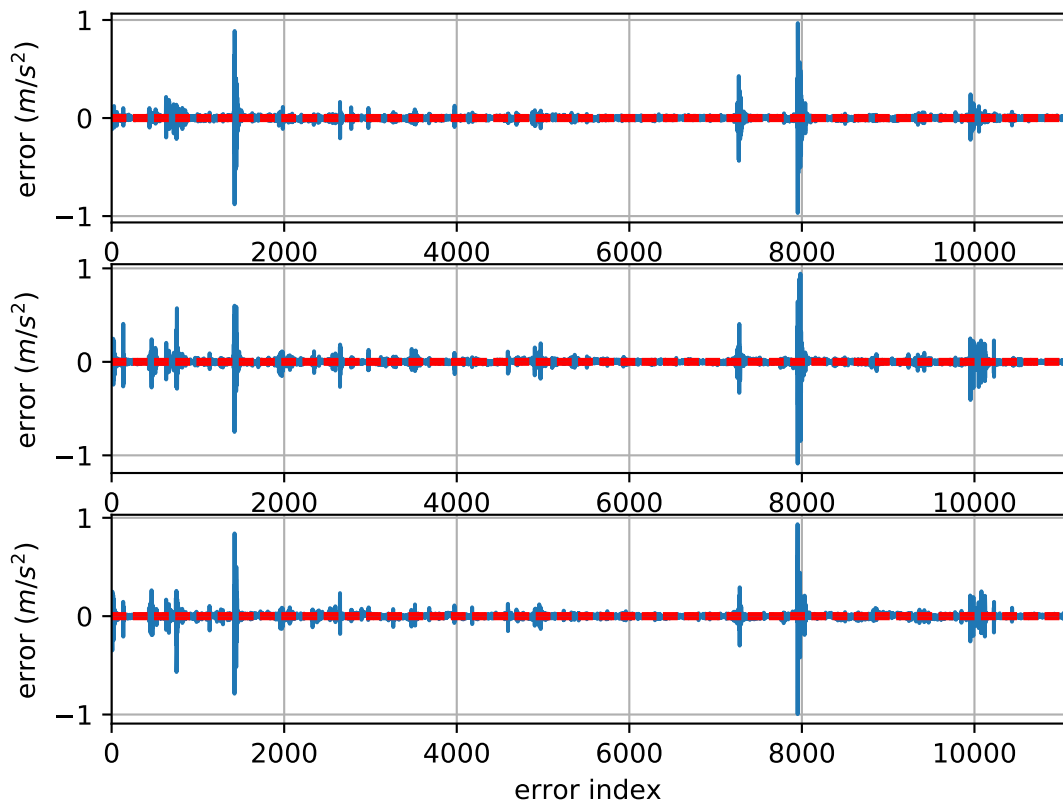
imu0: sample inertial rate



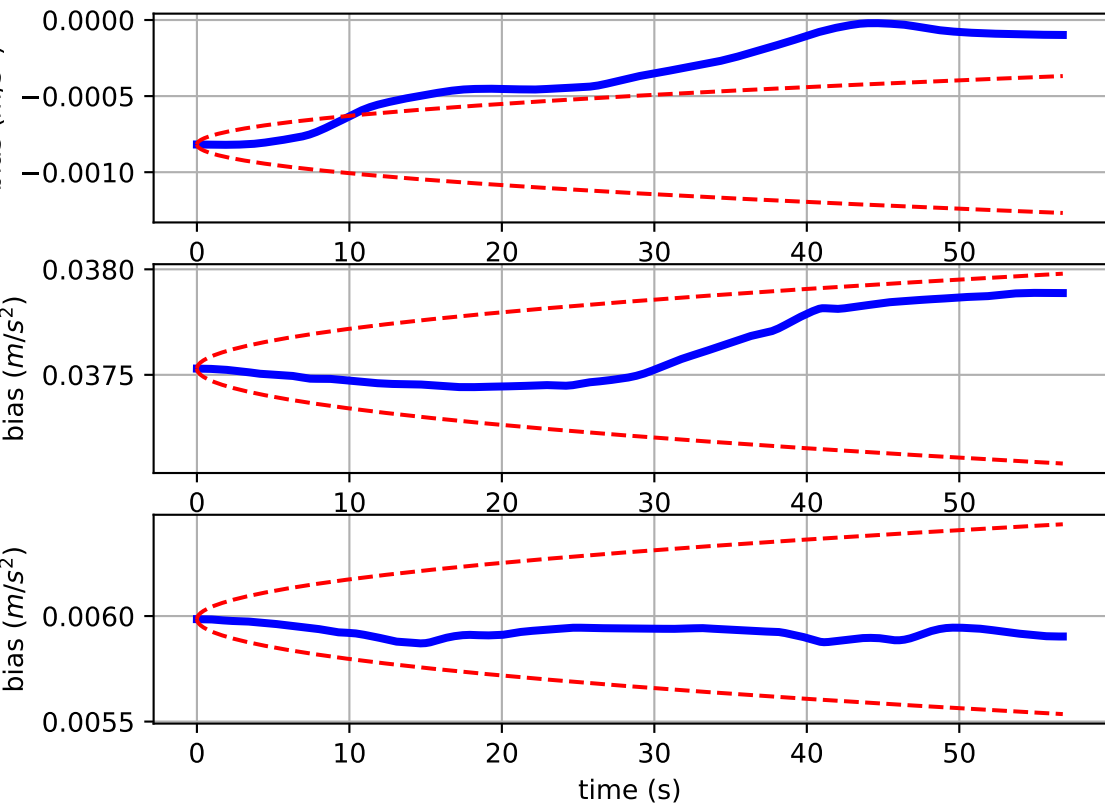
Comparison of predicted and measured specific force (imu0 frame)



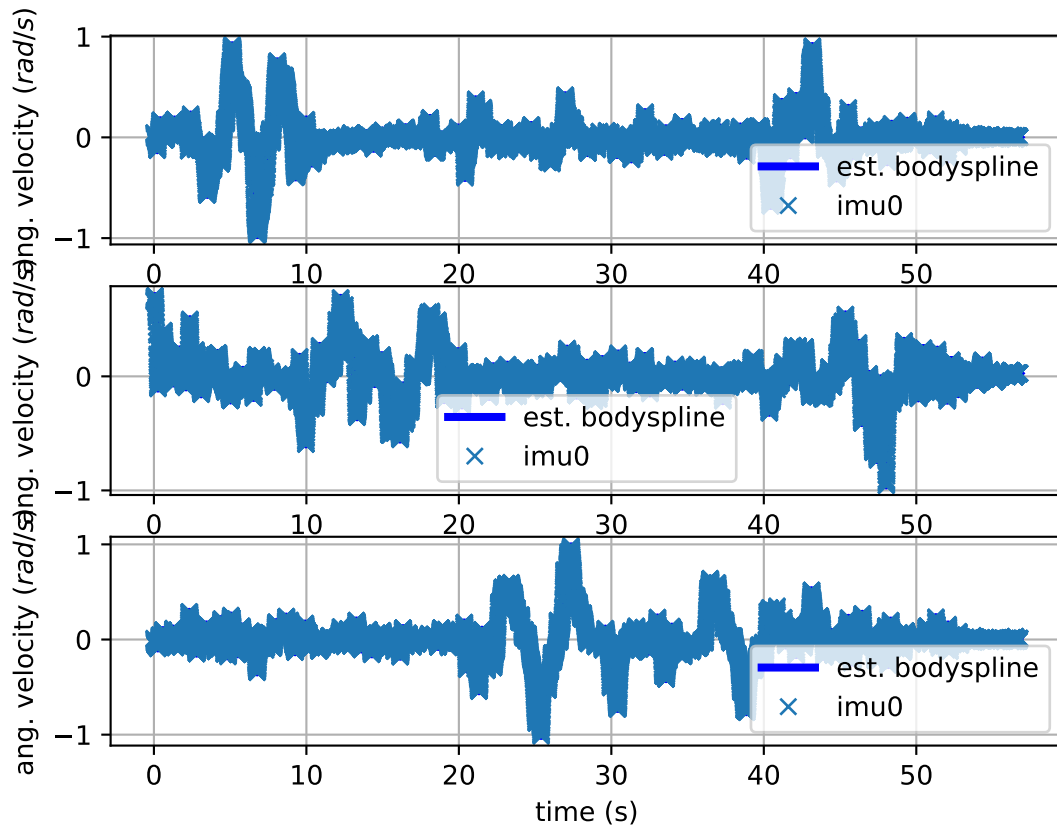
imu0: acceleration error



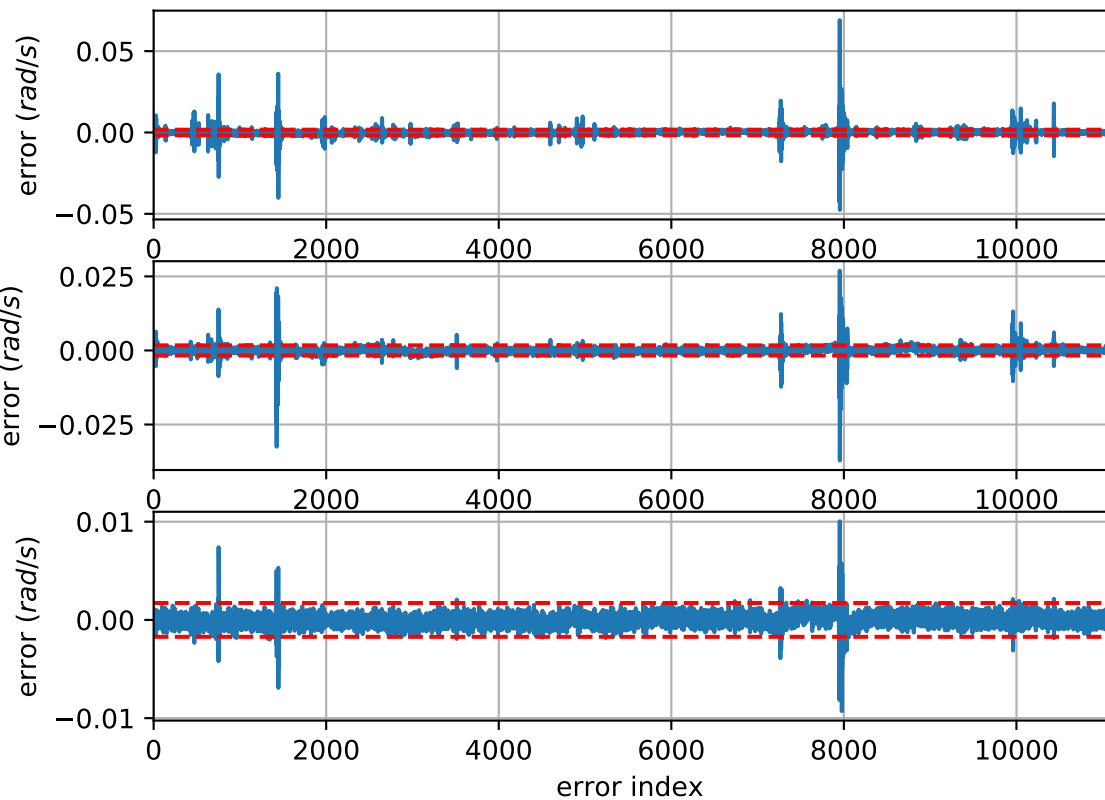
imu0: estimated accelerometer bias (imu frame)



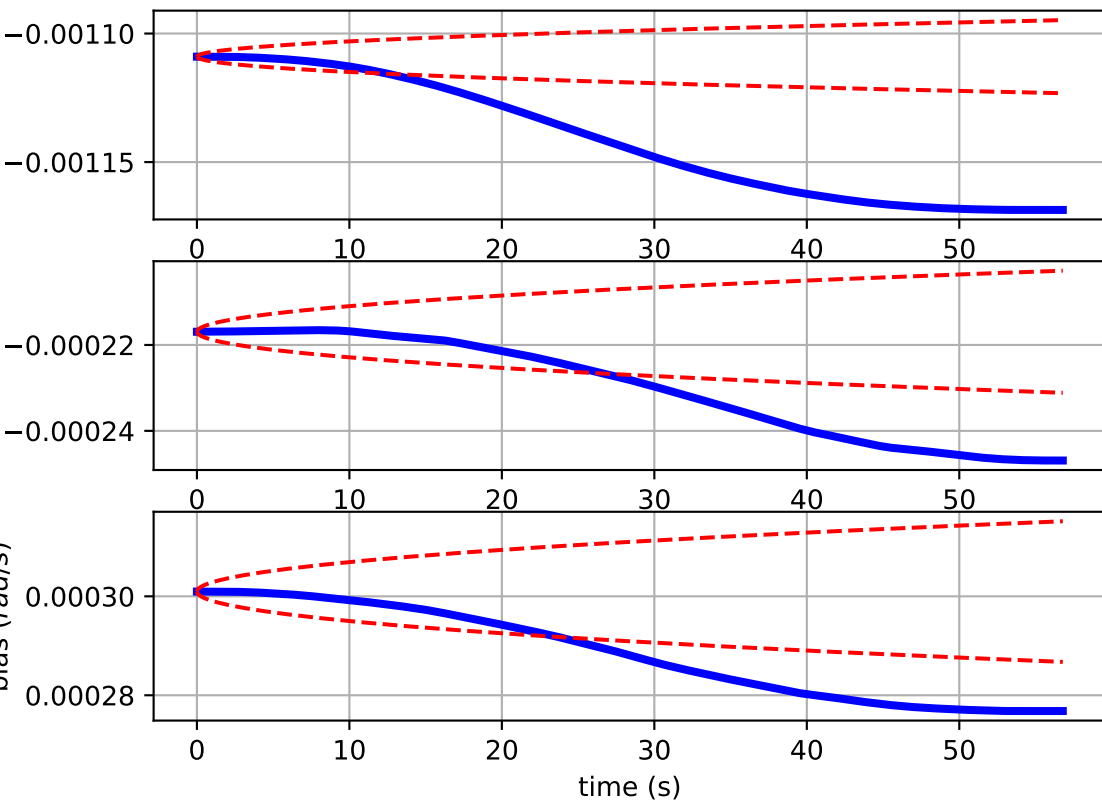
Comparison of predicted and measured angular velocities (body frame)



imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors

